

As Usual: $X^4 =$ closed, oriented 4-manifold

$P \rightarrow X =$ an $SU(2)$ - ^{G} or $SO(3)$ -Principal bundle

$\mathcal{A} = \{\text{Connections on } P\}$

$\mathcal{G} = \text{Aut}(P)$

$\mathcal{B} = \mathcal{A} / \mathcal{G}$

$\mathcal{A}^\Gamma = \{A \in \mathcal{A} \mid \text{stab}_{\mathcal{G}}(A) = \Gamma\}$

$\mathcal{B}^\Gamma = \mathcal{A}^\Gamma / \mathcal{G}$

$\mathcal{B}^* = \mathcal{B}^Z(G)$ "irreducibles"

New Notation: Let X have base pt x_0

$\mathcal{G}_0 = \{g \in \mathcal{G} : g(x_0) = 1\}$

$\tilde{\mathcal{B}} = \mathcal{A} / \mathcal{G}_0$

NB: $\mathcal{G}_0 \curvearrowright \mathcal{A}$ freely, since the only

constant gauge transformation is identity.

Big Picture: Understand topology of $\mathcal{B}^{(*)}$.

optionally

In particular, compute $H^*(\mathcal{B}^{(*)}; \mathbb{Q})$.

cohomology of

This will tell us about the moduli space of instantons.

Sketch: First compute $H^*(\tilde{B}; \mathbb{Q})$, then use Gysin sequence on base pt. fibration

$$\beta: \tilde{B}^* \longrightarrow B^*$$

$G_0 \curvearrowright A$ free $\Rightarrow \begin{array}{c} A \\ \downarrow \\ \tilde{B} \end{array}$ is a principle bundle. $\Rightarrow$ easier to compute cohomology

Recall, A affine over $\Omega^1(X; \text{Ad}(P))$ hence contractible.

Thus,

$$\tilde{B} \simeq_{\text{h.c.}} BG_0$$

Ignoring topological restrictions for a moment, for $G = SU(2)$,

$\downarrow$ base pt. preserving maps

$$\begin{aligned} BG_0 &= B\text{Map}_*(X, SU(2)) \\ &\simeq \text{Map}_*(X, BSU(2)) \end{aligned}$$

So $\tilde{B} \simeq \text{Map}_*(X, BSU(2))$.

Thm $\tilde{B} \simeq \text{Map}_* (X, BSU(2))_P$

↑ path component containing
the classifying map of P

Intuitively, $A \simeq \text{Hom}(TX, P)_P$,

on which G_0 acts freely, &
quotient yields result.

$BSU(2) = \mathbb{H}P^\infty$

NB: $H^*(BSU(2); \mathbb{Z}) \cong \mathbb{Z}[C_2],$
 $C_2 \in H^4(BSU(2); \mathbb{Z})$

A nice description of mapping space Cohomology

Defⁿ If A, B spaces
 $\varphi \in C^{p+q}(A \times B), \quad b \in C_q(B)$

The slant product is

↓ cross product of chains

$$\underbrace{(\varphi / b)}_{\in C^p(A)}(a) = \varphi(a \times b)$$

On de Rham cohomology, this is "fiber integration".

$$\omega \in \Omega^{p+q}(A \times B), \quad [F] \in H_q(B)$$

↑ Smooth rep

define

$$\underbrace{(\omega/[F])}_\alpha \in \underbrace{\Omega^p(A)}_{\Omega^p(A)} = \int_F \omega|_{\{a\} \times F}$$

A generalization
of the Fourier
transform.

In our case, $A = X$, $B = \text{Map}(X, Y)$.

Take $\varphi \in H^k(Y)$, so

$$\text{ev}^*(\varphi) \in H^k(X \times \text{Map}(X, Y))$$

So, for $a \in H_i(X)$,

$$\text{ev}^*(\varphi)/a \in H^{k-i}(\text{Map}(X, Y))$$

↑ called "transgressed classes"

∃ evaluation fibration

$$\text{Map}_*(X, Y) \rightarrow \text{Map}(X, Y) \rightarrow Y$$

↙ e.g. $BSU(2)$

If Y has "simple cohomology," Leray-Serre

Spectral sequence shows that $H^*(\text{Map}_*(X, Y))$

gen'd by transgressed classes.

Defⁿ There's a map $\tilde{\mu} : H_2(X) \rightarrow H^2(\tilde{B}; \mathbb{Z})$
 given by $a \mapsto \text{ev}^*(c_2) / a$
 $\cap$
 $\text{Map}_+(X, B\text{SU}(2))_P$

There's also a map $\mu : H_2(X) \rightarrow H^2(B^*; \mathbb{Q})$
 given by $a \mapsto -\frac{1}{4} \text{ev}^*(p_1) / a$
 $\cap$
 $\text{Map}(X, B\text{SO}(3))_P$

Thm Suppose $\pi_1(X) = 1$, and

$$H_2(X; \mathbb{Z}) = \mathbb{Z} \langle \Sigma_1, \dots, \Sigma_b \rangle$$

Then $H^*(\tilde{B}; \mathbb{Q})$ is a polynomial algebra gen'd
 by $\tilde{\mu}(\Sigma_1), \dots, \tilde{\mu}(\Sigma_b)$.

In particular, $H^{2n}(\tilde{B}; \mathbb{Q}) \cong \text{Sym}^n(H_2(X; \mathbb{Q}))$.

(when $\pi_1(X) \neq 1$, odd degrees are typically an exterior algebra in terms of $H_1(X)$)

Pf Since X is simply connected, it fits
 into a cofibration sequence (up to htpy)

$$\bigvee_1^b S^2 \hookrightarrow X \rightarrow S^4.$$

Applying $\text{Map}_*(-, BG)$ yields a fibration

$$\tilde{B}_{S^1, k} \rightarrow \tilde{B}_p \rightarrow \bigoplus_i^b \tilde{B}_{S^2}$$

Lem 1) $H^*(\tilde{B}_{S^2}; \mathbb{Q})$ is polynomial algebra
over $\tilde{\mathcal{M}}(\mathbb{C}S^2)$

$$2) H^i(\tilde{B}_{S^1}; \mathbb{Q}) = 0 \quad \forall i > 0.$$

Supposing the lemma, K nneth $\Rightarrow$

$$H^*(\tilde{B}_p; \mathbb{Q}) \cong \bigoplus_i^b H^*(\tilde{B}_{S^2}; \mathbb{Q})$$

$\Rightarrow$ Done $\checkmark$.

Will prove lemma if time. First, use result
to compute $H^*(B^*; \mathbb{Q})$.

$$\beta : \tilde{B}^* \rightarrow B^*$$

is a $\mathcal{G}/\mathcal{G}_0 \cong \text{SU}(2)/\mathbb{Z}(\text{SU}(2)) \cong \text{SO}(3)$ bundle.
 $\uparrow \{\pm 1\}$

So there's an additional class

$$v = -\frac{1}{4} p_1(\beta) \in H^4(B^*; \mathbb{Q}).$$

Thm $H^*(B^*; \mathbb{Q}) \cong \mathbb{Q}[v, \mu(\Sigma_1), \dots, \mu(\Sigma_b)]$

Pf: $\tilde{B}^* \simeq \tilde{B}$, since reducibles have infinite codimension in $\tilde{B}$.

$$H^*(SO(3); \mathbb{Q}) \cong H^*(\underset{S^3}{\underset{S^{11}}{SU(2)}; \mathbb{Q}),$$

So use Gysin sequence

$$\begin{aligned} \cdots \rightarrow H^{k-4}(B^*; \mathbb{Q}) &\xrightarrow{-\cup v} H^k(B^*; \mathbb{Q}) \xrightarrow{\beta^*} H^k(\tilde{B}^*; \mathbb{Q}) \\ &\xrightarrow{\delta} H^{k+1-4}(B^*; \mathbb{Q}) \rightarrow \cdots \end{aligned}$$

Since $\beta^*(\tilde{\mu}) = \mu$, β^* is surjective, so $\delta = 0$.

$\leadsto$ SES's

$$0 \rightarrow H^{n-4}(B^*; \mathbb{Q}) \rightarrow H^n(B^*; \mathbb{Q}) \rightarrow H^n(\tilde{B}; \mathbb{Q}) \rightarrow 0$$

$$\Rightarrow H^n(B^*) = H^n(\tilde{B}) \oplus (v \cup H^{n-4}(\tilde{B})) \oplus \dots$$

□

The μ map is essential in the definition of Donaldson invariants and will of course be relevant to orientation of the Moduli space.

Pf of lem: By previous observations,

$$\tilde{B}_{S^n} \cong \Omega^n(BSU(2)) = \text{Map}_*(S^n, \gamma)$$

↑ follows from Map_*
Smash adjunction.

Moreover,

$$\Omega^n BSU(2) \cong \Omega^{n-1} SU(2) = \Omega^{n-1} S^3$$

There's a fibration

path space



$$\Omega S^3 \rightarrow PS^3 \rightarrow S^3$$

Since PS^3 contractible, $E_{\infty}^{p,q} = 0$ in the Leray-Serre spectral sequence unless $p = q = 0$.

The cohomology of S^3 has non-trivial terms in degrees 0 & 3, hence d_3 is the only non-zero differential, & is an isomorphism except in degree 0.

$$E_2 = E_3 =$$

$\Rightarrow$

$$H^p(\Omega S^3; \mathbb{Q}) \cong \begin{cases} \mathbb{Q} & p \text{ even} \\ 0 & \text{else} \end{cases}$$

Leibniz rule of $d_3 \Rightarrow$ Powers of 2D classes $\neq 0$,
So $H^*(\Omega S^3; \mathbb{Q})$ is a polynomial algebra.

Similarly,

$$H^p(\Omega^2 S^3; \mathbb{Q}) \cong \begin{cases} \mathbb{Q} & p=0,1 \\ 0 & p \geq 2 \end{cases}$$

But for $\Omega^3 S^3$, problem.

$$\pi_0(\Omega^3 S^3) = \pi_1(\Omega^2 S^3) = \mathbb{Z}.$$

But from the spectral sequence, the cohomology of each component vanishes. (lift to universal cover of $\Omega^2 S^3$).

Now, Hurewicz $\Rightarrow$

$$H_2(\tilde{B}_{S^2}) = H_2(\Omega S^3) = \pi_2(\Omega S^3) = \pi_3(S^3) \cong \mathbb{Z}$$

Suffices to find a 2-sphere $T \subset \tilde{B}_{S^2}$ s.t.

$$\langle \tilde{\mu}([S^2]), [T] \rangle = 1.$$

This is equivalent to finding an $SU(2)$ bundle $\underline{P} \rightarrow S^2 \times S^2$ w/ $c_2(\underline{P}) = 1$, which is fine.

This shows that $\tilde{\mu}([S^2])$ generates $H^2(\tilde{B}_S^2)$,
so we're done.

□